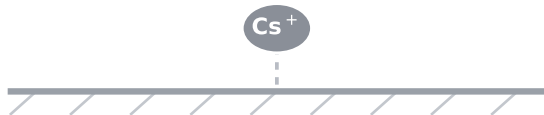


Cation uptake in geopolymers is ion exchange, not physisorption

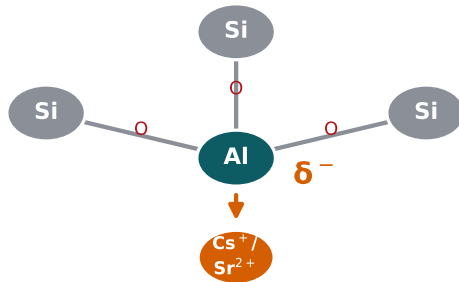
(a) physisorption on a surface



uptake $\propto$ surface area?

weak, non-specific

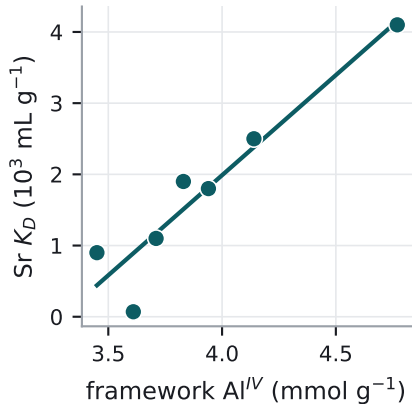
(b) ion exchange at framework Al^{IV}



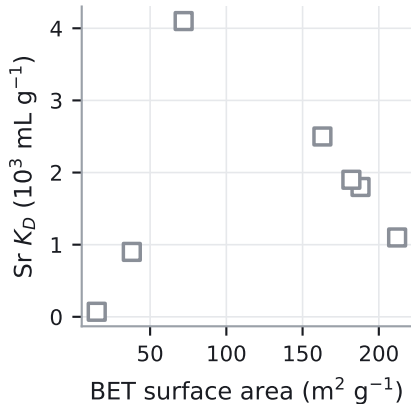
uptake $\propto$ framework Al^{IV}

strong, charge-balancing

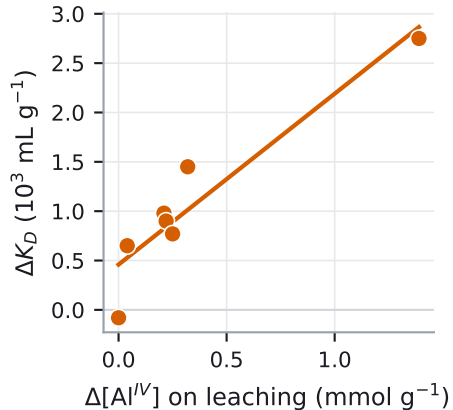
(a) framework Al
 $r = +0.95$



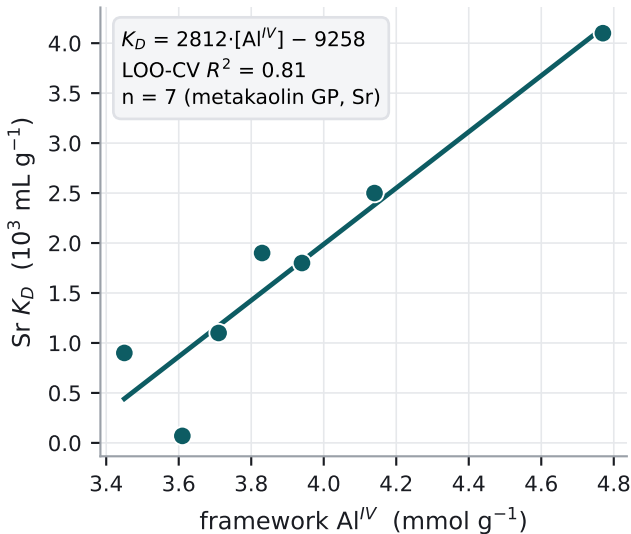
(b) surface area
 $r = +0.19$



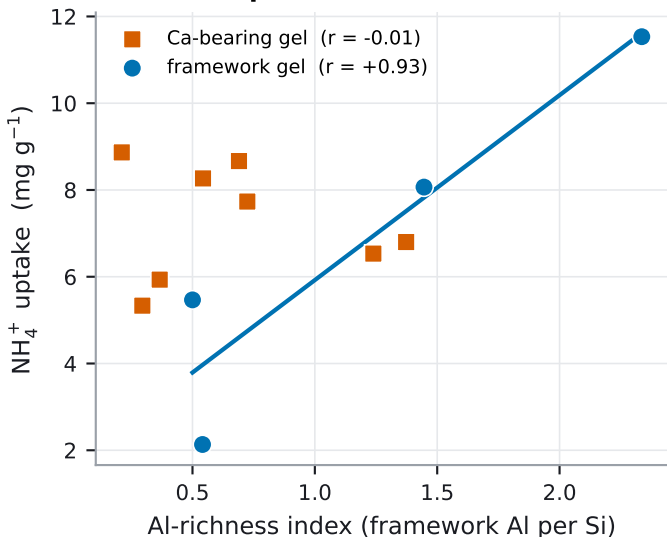
(c) causal test, n=7
 $r = +0.94$



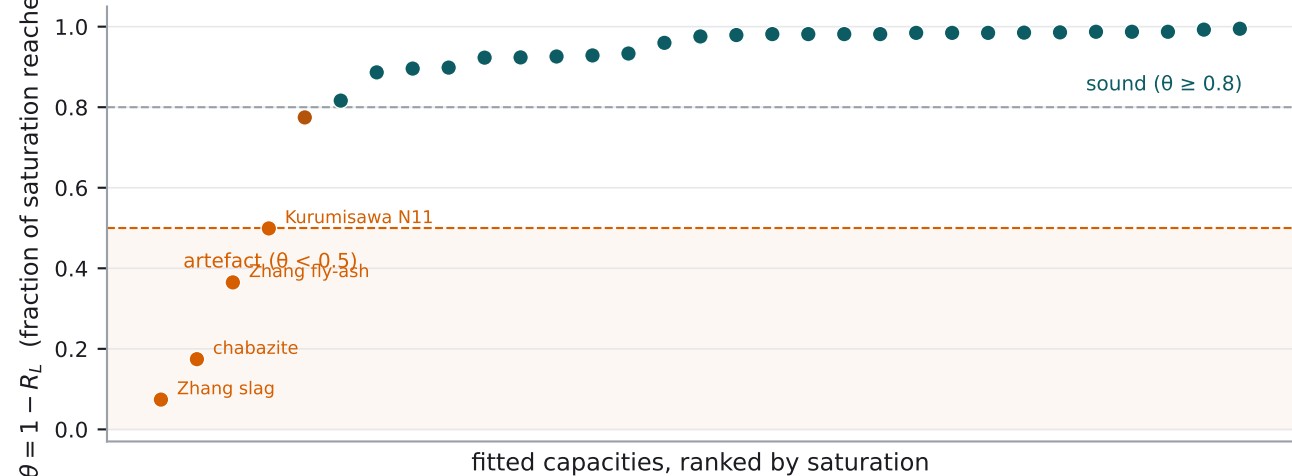
Within-class forward model



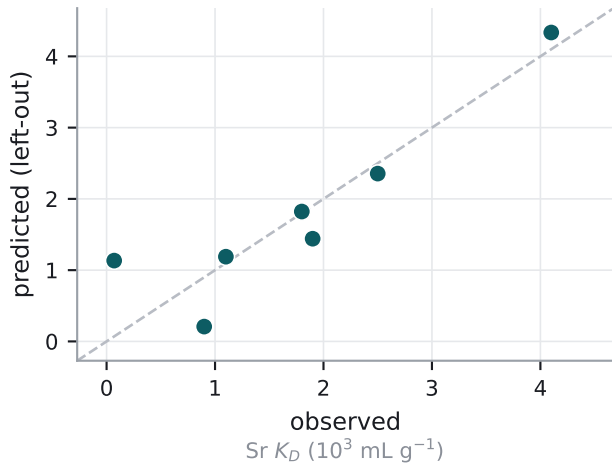
Structural precondition



Saturation screen: 4 of 31 capacities are extrapolation artefacts



(a) within one class / protocol
LOO-CV $R^2 = +0.81$ ✓ predictive



(b) pooled across laboratories
LOO-CV $R^2 = -0.09$ ✗ worse than the mean

